

The Order-2 FK term in the lensing two-point function: a numerical audit and corrected results

2026 August

Abstract

The Order-2 “FK” contribution plotted in the draft is not a faithful evaluation of the quantity the draft defines. The three-point driving-field vertex it needs is tabulated on a cosine mesh whose only samples on the relevant family are at zero separation and at 31° , so over the whole plotted range the vertex lookup returns the zero-separation value. The published curve’s angular dependence reproduces the interpolator’s weight function to one part in 10^4 , evidence internal to the deployed files. The amplitude at arcminute separations survives, to 0.15%; the shape does not, being overstated by factors of 5 at $22'$ and 100 at $114'$ and carrying the wrong sign beyond $230'$. Corrected, the FK term stays a small fraction of Order-0 at every separation instead of overtaking it near 2° , and it appears in ξ_- and $\xi_{\kappa\gamma}$ at a comparable level instead of vanishing.

Three further results are independent of that correction. The multipole cutoff of the vertex build is not converged: the tree-level integral turns over near $\ell \sim 10^4$, an order of magnitude above the deployed value, and carrying it there raises the arcminute amplitude from 0.5% of Order-0 to 2.31%. Replacing the tree-level bispectrum by BiHalofit does not converge at all, and not because the sum was stopped too early: the FK term samples a coincident-point moment, which is ultraviolet sensitive by construction and has a finite value only because the linear spectrum happens to be steep enough. With a nonlinear spectrum it has no limit, and the formalism supplies no smoothing scale to give it one, so the cutoff acts as an unstated regulator and the nonlinear FK term reaches 41% of Order-0. A finite beam width would supply the missing scale, and the algebra of that proposal is verified here; which scale to adopt is a choice about the observable that remains open. The absolute normalisation, by contrast, is correct: the pipeline’s own factor chain reduces to the standard Limber convergence bispectrum with ratio unity to machine precision. And the equal-split claim $\Delta C_\ell^{EE} = \Delta C_\ell^{BB}$ is a coincident-limit identity: away from coincidence the two polarizations separate and the E-mode comes to dominate, the B-mode falling from 0.95 of the E-mode power at $\ell = 60$ to 0.42 at $\ell = 1500$.

This note reports the measurements, the corrected figures, and the specific statements in the draft that do and do not survive. Section 13 records four findings that came later, while the revision was being carried out, and that changed what was done: the FK Monte-Carlo does not measure the FK diagram; the multi-redshift FF slices had never been computed; the multipole sum stops converging beyond about a degree, which bounds where FK may be quoted; and the ζ -slices figure was still drawn at the old cutoff.

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1 What was checked, and how

This note reports a numerical audit of the Order-2 “FK” contribution to the lensing two-point function, the term the draft plots in `figures/analysis3_NL0_FFFK.pdf` and `figures/analysis3_cl_full_vs_00.pdf` and quotes in `sections/insights.tex` and `sections/conclusion.tex`.

Six questions were asked, in this order.

1. Does the plotted FK curve evaluate the quantity the paper defines? (Section 2)
2. Is the multipole cutoff in the vertex build converged, and if not, what does the converged answer rest on? (Section 3)
3. Does the line-of-sight floor $\lambda_{\min}$ of the tabulated vertex bias the result, and does the bias depend on source redshift? (Section 4)
4. Is the absolute normalisation of the vertex right, having never been compared against anything external? (Section 7)
5. Do the draft’s statements about the E and B polarization structure survive a resolved vertex? (Section 8)
6. Does a nonlinear bispectrum give a more realistic prediction? (Section 9), and if the answer is no, what exactly is missing? (Section 10)

The first is the one that changes published numbers. The fourth is the one whose answer is unreservedly positive. The rest fall in between.

The first was settled by an independent reviewer who was given the figure production path and nothing else. That reviewer was not told what any earlier work had concluded, was blocked from reading the earlier notes, and was told explicitly that finding nothing wrong would be a useful result. It worked forward from the figure manifest through the generator scripts,

the folded `xi*.npz` files, the run configuration, the vertex callable and the tabulated vertex, testing rather than assuming at each step. Its conclusion agreed with the earlier work, and it added several results the earlier work did not have. The comparison is recorded in `notes/task_a_blind_audit_comparison.md`.

Two controls make the rest of the note interpretable.

The pipeline is reproducible. Re-running the production FK fold from a redirected copy of its own configuration reproduces the deployed result bit for bit, with maximum relative difference zero across all six component pairs. Every comparison below therefore isolates exactly one change.

The rebuild machinery is exact. The vertex table is rebuilt here in pieces: one radial-and-multipole LOW branch, and a set of HIGH multipole windows that are summed. This is only legitimate if the pieces are independent, so that was tested rather than assumed. Table rows are bit-exactly independent of one another in both branches: computing a row in a table of six rows and in a table of two rows gives identical doubles (`np.array_equal` true, maximum relative difference 0.000). The multipole windows are disjoint in maximum leg, so their sum is the same integral. Together these mean one build at the finest grid yields every coarser grid by row selection, and every smaller cutoff by partial summation, with no approximation. The reimplementations of the production triple mesh reproduces the deployed table’s 1671 rows exactly and in the same order.

All commands are in `driver_field_emulators/code/`, and every number quoted below names the script that produced it.

2 The vertex sampling problem

This section rests entirely on the deployed table and the deployed folded result. No rebuilt table is used, so none of it depends on trusting a second calculation.

2.1 What the fold asks for

The FK two-point fold evaluates the driving-field three-point cumulant on the collapsed configuration: two directions coincident, the third displaced by the plotted separation γ . In the cosine coordinates the vertex table is indexed by, that is the family $(1, \cos \gamma, \cos \gamma)$. This was established from the code rather than the documentation, by wrapping the production callable in a logging shim and running the production configuration. Across 1 990 656 recorded evaluations the fold requests exactly one sorted cosine triple per separation, and the three leg times agree to 2×10^{-13} , confirming the equal-time collapse. That is the object `sections/appendix.tex` defines in appendix *append: driving-field spectra*.

2.2 Where the table has samples

The deployed table is built on a uniform mesh of 15 cosines, spacing $1/7$, triangle filtered to 1671 rows, which the callable canonicalises to 351 unique sorted triples. On the collapsed family the mesh therefore has samples at

$$c = 1 \ (\gamma = 0), \quad c = 6/7 \ (\gamma = 31.0^\circ = 1860'), \quad c = 5/7 \ (\gamma = 44.4^\circ), \dots$$

There is no sample between $\gamma = 0$ and $\gamma = 1860'$. The figures plot $0.5'$ to $5000'$, that is 0.008° to 83° . Thirty-five of the forty plotted separations lie inside the first mesh gap.

The compression is worse than the gap width suggests, because the interpolation variable is $\cos \gamma$ and $1 - \cos \gamma \simeq \gamma^2/2$. Measured as a fraction of the first interval $[6/7, 1]$, the plotted

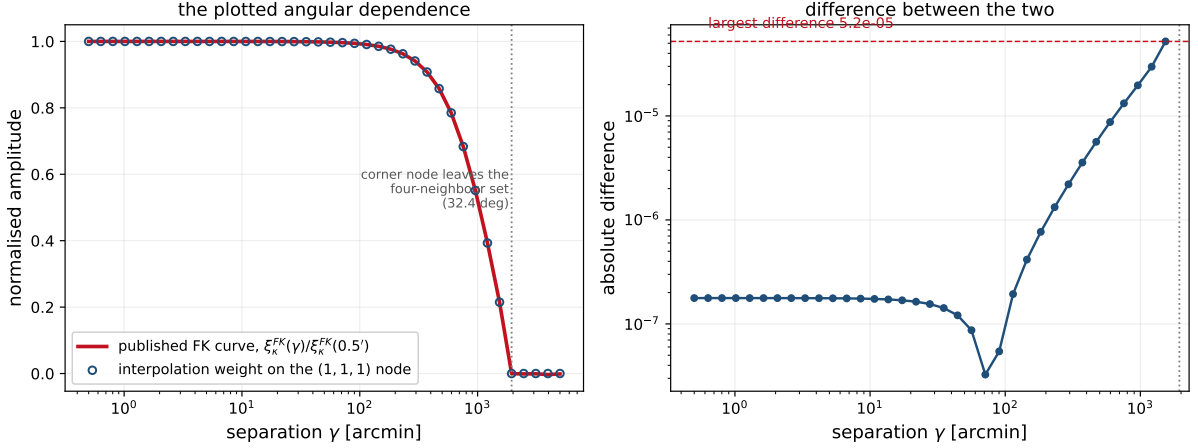


Figure 1: Left: the published FK curve for ξ_κ , normalised to its value at $\gamma = 0.5'$, against the normalised inverse-distance weight the vertex callable places on the $(1, 1, 1)$ node. Right: the difference between them. Both come from deployed artefacts only. The dotted line marks the separation at which the corner node leaves the four-neighbour set, which is where the published curve changes sign. Produced by `code/rebuild/probe_weight_identity.py`.

separations sit at 0.0000% ($\gamma = 0.5'$), 0.014% ($21.9'$), 0.62% ($144.7'$) and 4.1% ($372.2'$). Twenty-six of the forty are within 1% of the $c = 1$ endpoint. Both endpoints of the interval are exact table entries; the difficulty is that the plotted range does not span the interval, it hides inside one end of it.

2.3 The plotted curve is the interpolation weight

The callable looks a query up with a `cKDTree` over the 351 unique triples, taking the four nearest neighbours with inverse-distance weights. Reconstructing that lookup and reading off the normalised weight the rule places on the $(1, 1, 1)$ node gives a curve that can be compared directly with the published FK curve, normalised to its own value at $0.5'$.

They are the same curve. Over the whole range in which the corner node is in the neighbour set, $0.5'$ to $1535'$, the largest absolute difference is 5.2×10^{-5} and the largest relative difference is 2.4×10^{-4} (Figure 1). The plotted angular dependence of the FK term is the interpolator’s weight function. No property of the vertex enters it.

At $\gamma = 1944'$ the corner node leaves the four-neighbour set altogether, its weight becoming exactly zero, and the curve changes sign. That is 32.4° , and it is the feature the caption of Figure 15 in `sections/appendix.tex` describes as “the sign change at $\gamma \simeq 32^\circ$ ”. The four neighbours that replace it are $(1, \frac{6}{7}, \frac{6}{7})$, $(\frac{6}{7}, \frac{6}{7}, \frac{6}{7})$, $(1, \frac{5}{7}, \frac{5}{7})$ and $(\frac{6}{7}, \frac{6}{7}, \frac{5}{7})$: one collapsed configuration at 31° , one equilateral triangle and one scalene triangle. The sign change is a change of neighbour set.

2.4 The deployed table’s own next sample

The table also says, by itself, that the vertex cannot be flat across this range. On the source shell its collapsed-family entries read -1.35×10^{-14} at $c = 1$ and roughly three orders of magnitude smaller, with the opposite sign, at $c = 6/7$. The deployed file therefore records that the vertex falls by more than three orders of magnitude and changes sign somewhere inside the first gap. What it cannot record is where.

There is an independent reason to expect the decay, which needs no table at all. The collapsed configuration is the driving-field cumulant with two points coincident and the third displaced transversely by $r_\perp = \chi\gamma$ on the same shell. At the mid shell, $\gamma = 0.5'$ is $r_\perp \simeq 1$ Mpc

while $\gamma = 145'$ is $r_\perp \simeq 280$ Mpc. Matter correlations die well before 280 Mpc, so the vertex must decay across the plotted range. In the same units the mesh spacing of $1/7$ in $\cos \gamma$ is a resolution of about 3600 Mpc in $r_\perp$, while the physics happens between 1 and 100 Mpc.

2.5 The draft already contains the counterexample

Figure 7 of the draft, `zeta_driving_field_slices_draft.pdf`, is generated from `ell.band.decomp.results.npz`, which evaluates the vertex on the collapsed family at 28 separations directly, with no cosine mesh and no interpolation. It shows the vertex decaying steeply and changing sign inside the range over which the FK figures show it flat. The two figures in the same paper disagree, and the resolved one is right.

2.6 A second, smaller defect in the same callable

The callable sorts a query triple descending before lookup and then fills the coupling tensor with the retrieved value in every index permutation. Whether that is legitimate depends on the channel. Building the same collapsed configuration in the two leg orderings that canonicalise to the same sorted triple, and comparing, gives the median relative differences in Table 1.

channel	median relative difference	symmetric
ζ_{TTT}	3.3×10^{-3}	yes
ζ_{Bmod}	3.3×10^{-3}	yes
ζ_{PPP}	4.9×10^{-2}	no
ζ_{TPP}	2.3×10^{-1}	no
ζ_{TTP}	5.9×10^{-1}	no
ζ_{Dmod}	7.0×10^{-1}	no

Table 1: Sensitivity of each vertex channel to which leg carries the offset, over separations up to $200'$. The two orderings canonicalise to the same sorted triple, so the callable cannot distinguish them and the difference bounds the error its symmetric fill makes. Maxima are not quoted because every channel passes through zero in this range. Produced by `code/rebuild/probe_permutation.py`.

The channels that carry ξ_κ and ξ_+ are symmetric to the quadrature level, so the symmetric fill costs those two observables nothing. The channels that enter ξ_- and $\xi_{\kappa\gamma}$ are not, so those two can be quoted here as magnitudes and signs but not as precise values. This did not affect the deployed numbers, because the deployed lookup is pinned at $(1, 1, 1)$, which has no permutation duplicates. It becomes live exactly when the grid is refined, which is why it is fixed rather than merely noted: Section 6 reports the correction and what it changes.

3 The multipole cutoff

3.1 What the cutoff is

The vertex is built in two branches: an exact Wigner-3j sum for $\ell \leq 60$, and a flat-sky Limber integral above it, truncated at $\ell_{\max}$. The deployed value is $\ell_{\max} = 1000$. The question is whether the truncated integral has converged, and if not, what the converged answer depends on.

Sweeping the cutoff naively answers neither, for two reasons. A partial sum that is climbing slowly over one decade is indistinguishable from a divergent one. And the deployed quadrature maps a fixed number of Gauss-Legendre nodes linearly onto $[\ell_{\min}, \ell_{\max}]$, so raising the cutoff at fixed node count silently coarsens the rule and mixes truncation with a resolution artefact.

Both are avoided by integrating in octave windows, each with its own 96-node rule. The windows are disjoint in maximum leg, so their sum is the same integral, the node density per

octave is fixed by construction, and the partial sum up to any octave edge is the vertex built with that cutoff. One band set therefore gives the whole curve.

3.2 The integral converges, well above the deployed cutoff

Table 2 gives the per-octave contribution at the source shell, as a fraction of the total, together with the local slope p from fitting successive bands to L^{-p} . Positive p means the tail is dying.

band	$\gamma = 0.5'$			$\gamma = 21.9'$		
	contribution	fraction	p	contribution	fraction	p
(240, 480]	-4.19×10^{-15}	0.011	-3.04	-1.46×10^{-15}	0.162	-1.89
(480, 960]	-2.03×10^{-14}	0.055	-2.28	$+3.00 \times 10^{-16}$	-0.033	—
(960, 1920]	-5.57×10^{-14}	0.150	-1.46	-1.92×10^{-15}	0.213	—
(1920, 3840]	-9.69×10^{-14}	0.261	-0.80	-2.20×10^{-15}	0.244	-0.20
(3840, 7680]	-1.13×10^{-13}	0.303	-0.22	-1.96×10^{-15}	0.217	+0.17
(7680, 15360]	-8.14×10^{-14}	0.219	+0.47	-1.49×10^{-15}	0.166	+0.39

Table 2: Octave-band contributions to ζ_{TTT} at the source shell, tree level, on the corrected grid. The slope turns positive in the last band at both separations, so the integral converges; it does so near $\ell \sim 10^4$, an order of magnitude above the deployed cutoff. The top band still carries 17 to 22 percent of the total, so even $\ell_{\max} = 15360$ is short of the converged value by roughly that much. Produced by `code/rebuild/analyse.cutoff.py`.

So $\ell_{\max}$ is a numerical convergence parameter, not a physical smoothing scale, and the deployed value of 1000 sits on the rising side of the integrand, well below its peak. Relative to $\ell_{\max} = 960$, the vertex at $\gamma = 0.5'$ on the source shell grows by

$\ell_{\max}$	1920	3840	7680	15360
tree	3.24	7.14	11.67	14.95
BiHalofit	3.26	7.33	12.90	18.99

The two track each other to the first octave and then separate: by $\ell_{\max} = 15360$ the nonlinear vertex has grown by a further 27% relative to the tree one. That is the same statement as the slope table read the other way. The nonlinear spectrum is flatter, so its turnover is later, and its partial sum is still climbing where the tree one has begun to level off.

The mechanism is the effective spectral slope. After the radial collapse the hard pair contributes as a two-dimensional transverse integral $\int dk k P(k)$, whose integrand peaks where $n_{\text{eff}} = d \ln P / d \ln k$ equals -1 and which converges once $n_{\text{eff}} < -2$. The linear spectrum crosses -2 near $k \simeq 0.5 h/\text{Mpc}$, so the tree vertex turns over at multipoles of order 10^4 . The one-halo term flattens the nonlinear spectrum, pushing n_{eff} back up to about -1.1 at $k \simeq 1 h/\text{Mpc}$, so the nonlinear turnover is pushed much further out.

3.3 The observable, as a function of the cutoff

The vertex is not the observable. Folding each cutoff through the full pipeline gives Table 3.

This matters for how the draft’s headline number should be read. The 0.5% of Order-0 that the draft quotes at arcminute separations is the value at an unconverged cutoff. Carried to a cutoff where the integral has turned over, the tree-level FK term is closer to 2.5% of Order-0 there, a factor five above the quoted value. The grid correction of Section 2 and the cutoff correction therefore push in opposite directions: the first removes the wide-angle excess, the second raises the small-angle amplitude.

$\ell_{\max}$	$\xi_{\kappa}^{\text{FK}}(0.5')$	relative to $\ell_{\max} = 960$	as a fraction of Order-0
960	$+4.080 \times 10^{-6}$	1.00	0.48%
1920	$+8.160 \times 10^{-6}$	2.00	0.97%
3840	$+1.293 \times 10^{-5}$	3.17	1.53%
7680	$+1.708 \times 10^{-5}$	4.19	2.02%
15360	$+1.950 \times 10^{-5}$	4.78	2.31%

Table 3: The folded FK contribution to ξ_{κ} at the smallest plotted separation, tree level, on the corrected grid, against the multipole cutoff. Order-0 there is 8.443×10^{-4} . The successive factors are 2.00, 1.58, 1.32 and 1.14, so the observable is converging, and more quickly than the vertex does, because the fold weights the outer shells where the vertex converges first. Extrapolating the remaining increments puts the converged tree-level value near 2.5% of Order-0. Produced by `code/rebuild/sweep_cutoff.sh` and `code/rebuild/fig_cutoff.py`.

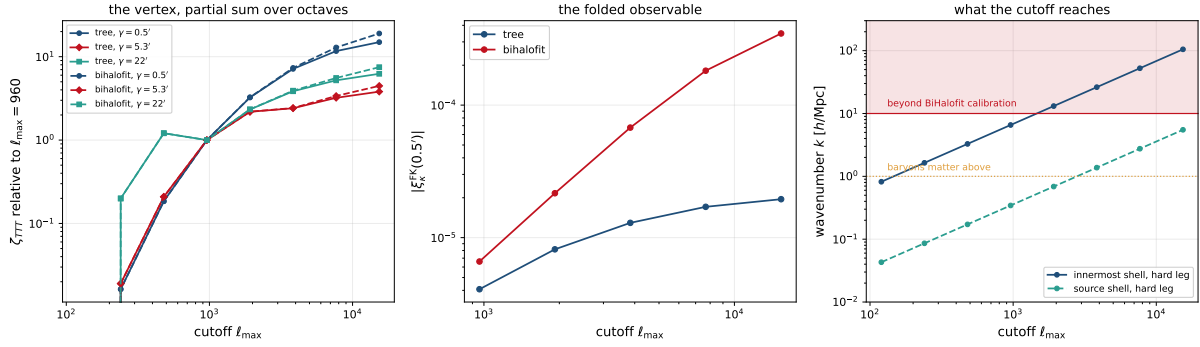


Figure 2: Left: the vertex as a partial sum over octave windows, normalised to the deployed cutoff. Middle: the folded observable at $\gamma = 0.5'$. Right: the wavenumber the hard leg reaches at the innermost and outermost shells, against the range where the nonlinear bispectrum fit is calibrated (shaded) and where baryons stop being a small correction.

3.4 What a converged value would rest on

This is the part that matters more than the cutoff itself. Under the Limber branch a multipole ℓ at a shell of comoving distance χ_h carries $k = \ell/\chi_h$, and the hard leg reaches $k = 2\ell/\chi_h$. Table 4 gives both at the innermost and outermost shells.

$\ell_{\max}$	innermost shell, $\chi_h = 293 \text{ Mpc}/h$		source shell, $\chi_h = 5582 \text{ Mpc}/h$	
	soft k	hard k	soft k	hard k
960	3.28	6.56	0.17	0.34
1920	6.56	13.13	0.34	0.69
3840	13.13	26.25	0.69	1.38
7680	26.25	52.50	1.38	2.75
15360	52.50	105.00	2.75	5.50

Table 4: Wavenumbers in h/Mpc reached at each cutoff. BiHalofit is fitted to simulations resolving $k \lesssim 10 h/\text{Mpc}$; baryonic feedback stops being a small correction above $k \sim 1 h/\text{Mpc}$.

At the deployed cutoff the hard leg reaches $6.6 h/\text{Mpc}$ at the innermost shell, inside the range where a nonlinear fit is calibrated. At the cutoff where the integral has actually turned over it reaches 50 to $100 h/\text{Mpc}$, which is inside single haloes: an order of magnitude beyond BiHalofit's calibration, and a regime where baryonic physics rather than gravity sets the matter spectrum.

The honest statement is therefore not “the FK amplitude is X ”. It is that the tree-level FK amplitude converges by $\ell_{\max}$ of order 10^4 , and that a converged *nonlinear* FK amplitude is a statement about the one-halo regime rather than about the bispectrum physics the draft describes. Reporting the amplitude as a function of the small-scale cutoff is the defensible presentation.

3.5 A practical ceiling

The Limber branch evaluates the hard leg at $k = 2\ell_{\max}/\chi_h$, and the fiducial $P(k)$ table stops at $k_{\max} = 206 h/\text{Mpc}$, with no extrapolation. At the innermost deployed shell, $\chi_h = 293 \text{ Mpc}/h$, that caps the reachable cutoff at $\ell_{\max} = k_{\max}\chi_h/2 \simeq 30\,000$. The cutoffs used here stay inside that. Pushing further needs a longer $P(k)$ table, and the same relation read the other way is the structural floor on the smallest usable shell discussed in Section 4.

4 The line-of-sight floor

4.1 What is cut, and why it depends on source redshift

The vertex table’s shells begin at $\lambda = 396.63 \text{ Mpc}$ and the callable returns exactly zero outside them, so everything nearer than that contributes nothing to the fold. Because λ is measured from the observer, that floor always corresponds to $z = 0.100$, whatever the source redshift: what is excised is always the $z < 0.1$ foreground.

The production fold uses a 24-node Gauss-Legendre rule whose nodes span $[5.6, 2307.5]$ Mpc, six of which fall below the floor. By plain measure that is 15% of the integration range, but plain measure is the wrong weight: the lensing efficiency $K = a^2\chi(\chi_s - \chi)/\chi_s$ vanishes toward the observer, and the FK term carries four response lines, so K^2 and K^4 bracket the weighting. Both fractions grow as the source moves nearer, which is the reason this matters for the multi-redshift figure and not for the main analysis.

z_s	λ_s [Mpc]	path fraction	K^2 weight	K^4 weight
5.0	2318.0	0.171	3.08%	0.67%
3.2	2266.6	0.175	3.47%	0.82%
2.5	2216.8	0.179	3.81%	0.96%
1.7	2094.9	0.189	4.64%	1.34%
1.0	1822.7	0.218	7.01%	2.62%
0.5	1330.7	0.298	15.95%	9.52%

Table 5: Kernel weight carried by the excised foreground, as a function of source redshift, with $K = a^2\chi(\chi_s - \chi)/\chi_s$ and the integral taken in the affine parameter on the same background the fold uses. The K^4 column is the relevant bracket for a term with four response lines. Both columns are upper bounds, because they ignore the vertex itself being some five orders of magnitude smaller at the innermost covered shell than at the outermost. Recomputed here rather than carried over; an earlier estimate in `notes/finding_lambda_min_cut.md` used a different weighting and gave figures about half these.

4.2 A structural floor, separate from the inherited one

The deployed floor of 396.63 Mpc is not a physical or numerical requirement. It is inherited from the shared radial grid the two-point and three-point tables both use. There is, however, a genuine floor underneath it: the Limber branch evaluates the hard leg at $k = 2\ell_{\max}/\chi_h$, so as a shell approaches the observer the required wavenumber diverges. With the fiducial $P(k)$ table

stopping at $k = 206 h/\text{Mpc}$, the innermost usable shell is

$$\chi_h = \frac{2\ell_{\max}}{k_{\max}} = 9.7 \text{ Mpc}/h, \quad \chi = 14.5 \text{ Mpc} \quad \text{at } \ell_{\max} = 1000.$$

A build at $\lambda = 6 \text{ Mpc}$ fails loudly with a request above $k_{\max}$, which is the correct behaviour: the code refuses to extrapolate $P(k)$ silently. Two consequences follow. The deployed floor is far above this structural one and can be lowered without touching the $P(k)$ table. And the cutoff and the smallest reachable shell are coupled: raising $\ell_{\max}$ raises the structural floor in proportion, so the convergence study of Section 3 and this one are not independent.

4.3 The floor measured, across source redshift

The estimate above is an upper bound. What it costs in practice is measured by rebuilding the vertex on a radial grid extended from the deployed floor of 396.6 Mpc down to 50 Mpc, four extra shells, and folding both tables at each source distance. The sixteen shared shells agree to machine zero between the two builds, so the extra shells perturb nothing and the difference is the floor alone.

z_s	median ratio, extended over standard	range over $\gamma \leq 200'$
5.0	1.0014	1.0011 to 1.0901
4.0	1.0017	1.0013 to 1.0711
3.2	1.0021	1.0016 to 1.0607
2.5	1.0028	1.0022 to 1.0561
1.7	1.0049	1.0039 to 1.0562
1.0	1.0127	1.0107 to 1.0783
0.5	1.0683	1.0610 to 1.2060

Table 6: What the $\lambda_{\min}$ floor costs the folded FK term, by source redshift. The upper end of each range is reached near the sign change, where the absolute values are small. Produced by `code/rebuild/fig_redshift.py`.

At the paper’s own source redshift the floor is worth 0.14%, so it affects nothing in the main analysis. At $z_s = 0.5$, the lowest slice of the multi-redshift figure, it is worth 6.8%. The measured numbers sit comfortably below the K^4 bracket of Table 5, as they should, since that bracket ignores the vertex being five orders of magnitude smaller near the observer.

This is the first time the floor has been measured at low source redshift, and it is the effect that matters for the multi-redshift figure rather than for the main result. The remedy is cheap: the deployed floor is inherited from a shared radial grid, not required by anything, and lowering it costs one extra vertex build.

4.4 The redshift trend, on a resolved vertex

The draft states that the FK leakage “strengthens steadily with source redshift”. That claim has never been measured on a resolved vertex, and two separate effects bias it in exactly that direction: this floor, whose kernel weight grows toward low z_s , and the cosine sampling, whose severity depends on the angular range each slice covers. Measuring it now is straightforward, because the propagator tables are built to $t_{\max}$ independently of the source distance, so a list of source distances costs one sweep rather than one run each.

Table 7 gives the corrected FK contribution relative to Order-0, at two separations.

The trend is weaker and less simple than the draft’s phrasing suggests. At arcminute separations the ratio is not monotonic: it rises from 0.40% at $z_s = 0.5$ to a maximum of 0.55% near $z_s = 1.7$ and then falls slowly back to 0.50% at $z_s = 5$. At 21.9’ it does rise monotonically,

z_s	λ_s [Mpc]	FK/Order-0 at 0.5'	FK/Order-0 at 21.9'
0.5	1330.7	0.00404	0.00387
1.0	1822.7	0.00524	0.00546
1.7	2095.0	0.00547	0.00628
2.5	2216.8	0.00544	0.00663
3.2	2266.6	0.00525	0.00671
4.0	2297.3	0.00516	0.00680
5.0	2318.0	0.00503	0.00683

Table 7: The corrected FK term relative to Order-0 against source redshift, at fixed cutoff, on the corrected grid and with the corrected callable of Section 6. Produced by `code/rebuild/fig_redshift.py`.

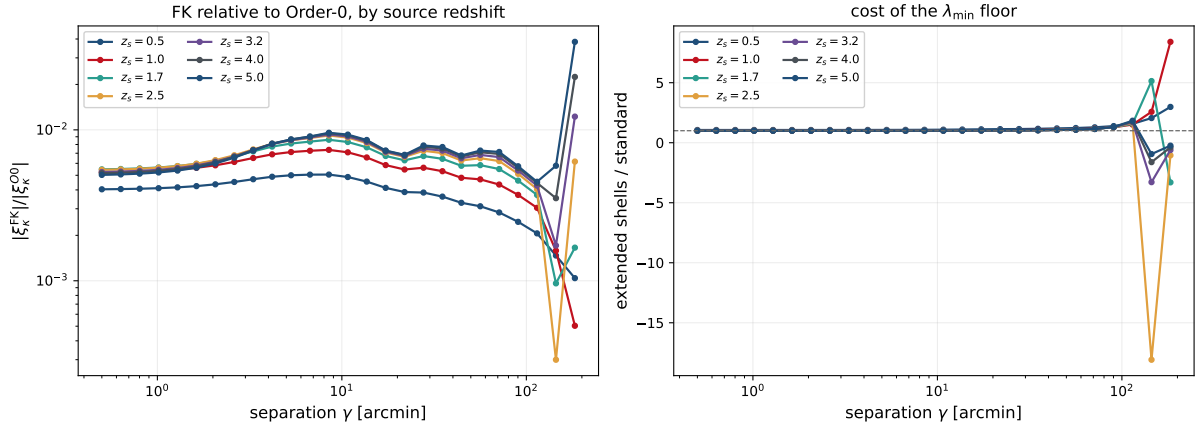


Figure 3: Left: the corrected FK contribution relative to Order-0 against separation, at each source redshift. Right: the same FK calculation with the tabulated shells extended from the deployed floor of 396.6 Mpc down to 50 Mpc, divided by the standard one. The right panel is what the $\lambda_{\min}$ floor costs, and it is largest at low source redshift because the kernel weight of the excised foreground grows as the source moves nearer. Produced by `code/rebuild/fig_redshift.py`.

but by a factor 1.7 across the whole range rather than steadily and strongly. The absolute FK amplitude does grow sharply with source redshift, by a factor 92 at 0.5' between $z_s = 0.5$ and $z_s = 5$, but so does Order-0, and it is the ratio that carries the claim.

So the statement should be weakened rather than removed: FK relative to the linear signal is largest at intermediate to high source redshift, and depends on separation as well as on z_s .

The two effects can now be separated, which was the point of measuring both. Correcting each slice for its own $\lambda_{\min}$ bias using Table 6 raises the $z_s = 0.5$ point by 6.8% and the $z_s = 5$ point by 0.14%, so the ratio between the extremes falls from 1.77 to 1.65 at 21.9' and from 1.25 to 1.17 at 0.5'. The floor therefore accounts for a small part of the apparent trend and not most of it. The larger part of the draft's overstatement came from the cosine sampling, which is corrected here throughout.

5 Corrected results

5.1 What was rebuilt

The corrected vertex table carries the 1671 production triangle-filtered rows unchanged, so it still serves the open-triangle consumers, and adds the collapsed family $(1, \cos \gamma, \cos \gamma)$ sampled at the forty plotted separations and refined between them. Three densities were built, with 40, 79 and 157 collapsed nodes, by row selection from a single build, which is exact because rows

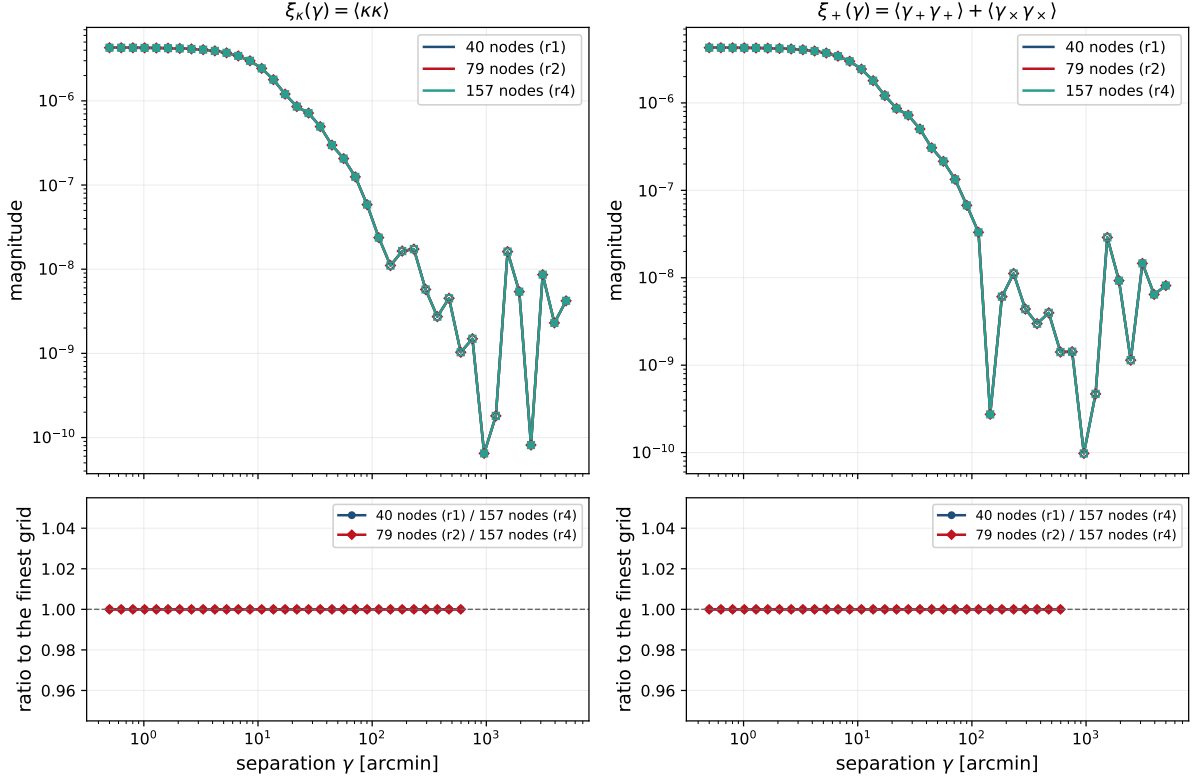


Figure 4: Folded FK curves at three collapsed-grid densities, and their ratios to the finest. The three agree to 4×10^{-4} . Produced by `code/rebuild/fig-convergence.py`.

are independent.

Everything else is held at the deployed values: the same sixteen radial shells, the same cosmology, the same tree-level bispectrum, the same multipole cutoff of 1000, the same single quadrature window with 96 Gauss-Legendre nodes, and the same fold. Assembled on the deployed rows and settings, the rebuild reproduces the deployed table to a median relative difference of 3×10^{-16} across all six channels, that is to round-off. The difference between the deployed and corrected curves is therefore the cosine sampling and nothing else.

5.2 The grid density has converged

Refining the collapsed family from 40 to 157 nodes leaves all four folded observables *bit-identical* (Figure 4). This is stronger than convergence and it has a clean reason: once the queried configurations are tabulated and the mesh key is fine enough to recognise them, the callable returns the tabulated row directly and never consults the refinement rows, so density beyond the queried set provably cannot matter. With the production callable’s coarser key the same comparison gave differences of up to 4×10^{-4} , because thirty-eight of the forty queries were failing the exact-match test and being interpolated unnecessarily; Section 6 measures that.

The corrected curve is therefore a property of the vertex and not of the mesh, in the strongest sense available: it is what the tabulated rows say.

5.3 The corrected two-point curves

Table 8 gives the numbers for ξ_κ .

Three things change and one does not.

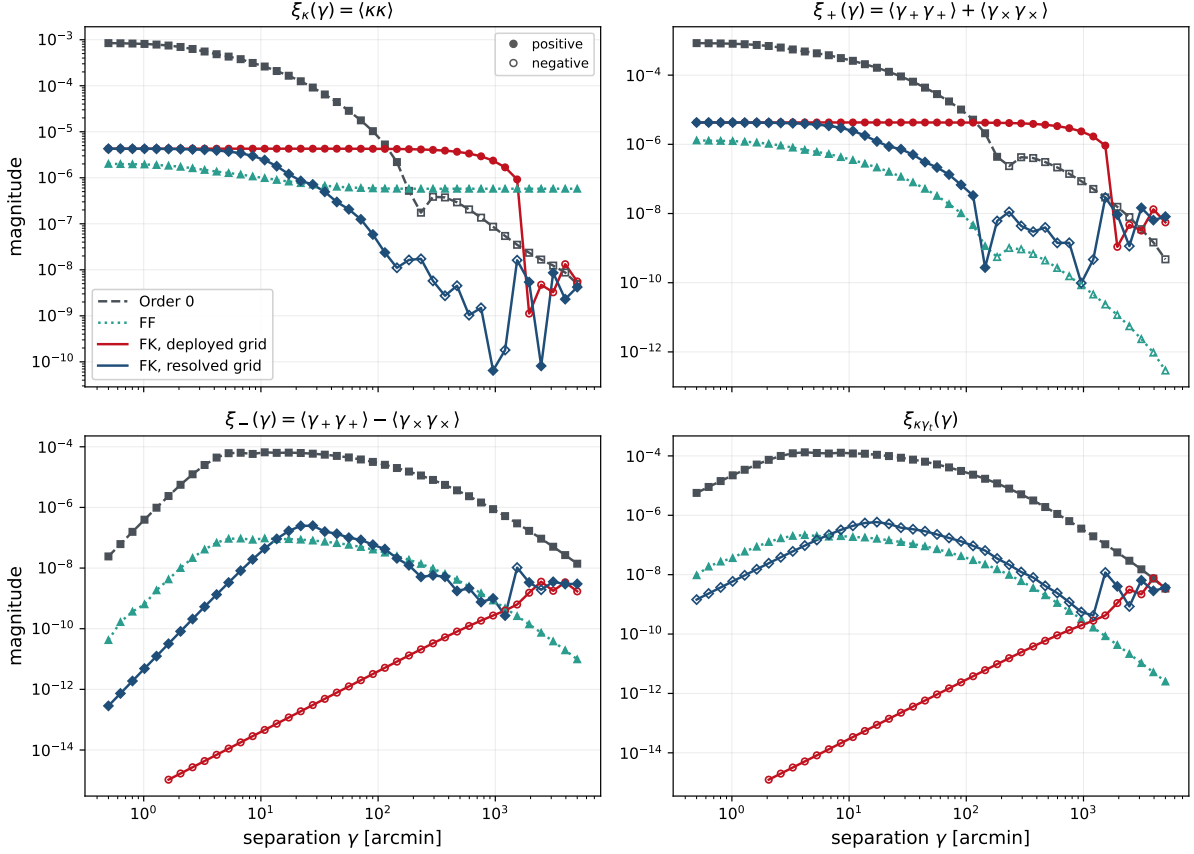


Figure 5: The four plotted observables, with Order-0 and the Order-2 FF term for scale. Red is the deployed FK curve, blue the same calculation with the vertex sampled on the family it queries. Filled markers are positive, open markers negative. Produced by `code/rebuild/fig-grid-comparison.py`.

The arcminute amplitude survives. At $\gamma = 0.5'$ the corrected value is $+4.2815 \times 10^{-6}$ against the published $+4.2877 \times 10^{-6}$, a difference of 0.15%. The queried configuration there really does sit on the tabulated corner, so the corner value is the right answer. Every statement the draft makes about the arcminute amplitude, including that FK is about twice FF there, stands.

The shape does not. By $22'$ the published curve is too large by a factor 4.9, by $114'$ by a factor 100, and beyond about $230'$ it has the wrong sign as well.

There is no crossover. At the deployed cutoff the corrected FK/Order-0 ratio is between 0.5% and 1% at every separation that carries signal. The two terms decay together. The published crossover near $145'$ was the frozen corner value being compared against an Order-0 term that genuinely decays. The ratio itself is cutoff dependent, rising to between 1.4% and 2.3% at $\ell_{\max} = 15360$ (Section 3); what does not depend on the cutoff is that it stays well below unity at every separation, so no crossover appears at any cutoff tested.

The spin-2 observables are not empty. With the vertex resolved, ξ_- reaches $+2.50 \times 10^{-7}$ near $28'$ and $\xi_{\kappa\gamma_t}$ reaches -5.89×10^{-7} near $17'$, about 0.4 and 0.5% of Order-0, where the deployed run puts them at 10^{-16} to 10^{-9} . The exact cancellation the draft reports is a $\gamma \rightarrow 0$ statement: at zero separation ζ_{TPP} , ζ_{PPP} and ζ_{TTP} vanish and the ξ_- combination cancels identically. It is not a statement about the plotted range. These two values are quoted after the callable corrections of Section 6; without them they would read $+2.86 \times 10^{-7}$ and -8.88×10^{-7} .

γ [arcmin]	FK deployed	FK corrected	deployed/corrected	corrected/Order-0
0.50	$+4.2877 \times 10^{-6}$	$+4.2815 \times 10^{-6}$	1.00	0.0051
5.30	$+4.2876 \times 10^{-6}$	$+3.7270 \times 10^{-6}$	1.15	0.0087
8.51	$+4.2875 \times 10^{-6}$	$+3.0037 \times 10^{-6}$	1.43	0.0096
21.88	$+4.2863 \times 10^{-6}$	$+8.7337 \times 10^{-7}$	4.91	0.0070
44.43	$+4.2817 \times 10^{-6}$	$+3.1491 \times 10^{-7}$	13.60	0.0071
114.27	$+4.2483 \times 10^{-6}$	$+4.2402 \times 10^{-8}$	100.19	0.0080
183.26	$+4.1873 \times 10^{-6}$	$+4.1908 \times 10^{-9}$	999.15	0.0081
232.08	$+4.1282 \times 10^{-6}$	-5.1152×10^{-9}	-807	0.0293
596.89	$+3.3655 \times 10^{-6}$	-1.8661×10^{-9}	-1803	0.0091

Table 8: The FK contribution to ξ_κ before and after resolving the vertex, at fixed everything else. Beyond about $230'$ the corrected curve has changed sign, so the ratio is negative and its magnitude is not meaningful. Full table in `products/grid-comparison-table.txt`.

5.4 Harmonic space

Transforming with the draft’s own curved-sky Wigner-d kernels gives Table 9.

ℓ	$C_\ell^{\kappa\kappa}$, Order-0	FK/Order-0, deployed	FK/Order-0, corrected
3	6.34×10^{-8}	+17.95	-0.197
5	7.97×10^{-8}	+9.02	-0.002
7	8.95×10^{-8}	+3.81	+0.084
9	9.75×10^{-8}	+0.87	+0.034
39	9.86×10^{-8}	+0.0008	+0.0088
115	4.61×10^{-8}	-0.0012	+0.0079
415	8.33×10^{-9}	-0.0001	+0.0091
789	2.52×10^{-9}	-0.0001	+0.0089

Table 9: FK relative to Order-0 in $C_\ell^{\kappa\kappa}$, using the generator’s own transform. The factor 18 excess at $\ell = 3$ is the sampling artefact. The corrected FK is a steady 0.8 to 0.9% correction over $39 \leq \ell \leq 800$; the residuals at $\ell < 10$ are of order 10 to 20 percent and are within the finite-range and apodisation systematics the generator documents. Produced by `code/rebuild/cl_ratio.py`.

6 The two callable defects, fixed

Section 2 identified two defects in the vertex callable alongside the sampling problem, and earlier drafts of this note quoted ξ_- and $\xi_{\kappa\gamma_t}$ only as magnitudes because of the first of them. Both are now corrected, in `code/callable.fixed/perm_aware_kappa3_callable.py`, and this section reports what changes. The production callable is not overwritten; the replacement is exercised through the production runner with `run_fk_variant.py --callable`, so the fold is unchanged.

6.1 Leg permutations

The tabulated channels fix which leg carries which field: ζ_{TPP} has spin weights $(0, 0, 2)$, so the spin-2 leg is the third, and ζ_{TPP} has $(0, 2, 2)$, so the scalar leg is the first. The production callable sorts the query triple and then writes one value into every index permutation of the coupling tensor. Asking for the spin field on a different leg is a question about a *permuted geometry*, not about the same geometry with a relabelled channel, so the symmetrised fill is

exact only for the fully symmetric ζ_{TTT} and for geometries that happen to be permutation symmetric.

The correction evaluates each entry at the geometry its own field assignment requires. Four of the five cases are a single lookup at a permuted triple. The fifth, three spin legs with one Re and two Im, is the case the table cannot answer with one lookup: $(\zeta_{Dmod} - \zeta_{PPP})/4$ is the *average* of two of the three placements of the Re leg, so three cyclically permuted evaluations are combined to isolate the wanted one. On a symmetric geometry the three coincide and the combination collapses to the production value.

Two consequences are worth stating before the numbers. The corrected lookup asks for permuted geometries, so the table must contain them; the production mesh already does, and the densified collapsed family is made so by `code/rebuild/make_perm_closed_triples.py`. And the callable now checks that closure at load, warning rather than letting a missing row be answered silently by a different configuration. The deployed table is 99.8% closed, the four exceptions being fully degenerate rows on the triangle-inequality boundary that no fold queries.

6.2 The mesh key

The production callable deduplicates triples after rounding to eight decimals, so two collapsed configurations closer than $1 - \cos \gamma = 5 \times 10^{-9}$ become one node however finely the table samples. That is a floor at $\gamma = 0.344'$, against a smallest plotted separation of $0.5'$. The key is now twelve decimals, a floor at $0.0034'$. The production mesh deduplicates to the same 351 rows at either precision, so no coarse-grid behaviour changes.

The floor was not the whole cost, and the rest of it was not anticipated. The callable has a fast path: if the query lands on a tabulated row to within 10^{-10} it returns that row directly instead of blending four neighbours. Measuring the nearest-neighbour distance for the forty plotted separations against the refined table shows that with the eight-decimal key only *two* of the forty are exact hits, the rest missing by up to 7×10^{-9} , purely because the key had rounded the row away from the query. With twelve decimals all forty are exact hits, the largest residual being 7×10^{-13} .

So the coarse key was forcing an unnecessary four-neighbour interpolation at thirty-eight of the forty separations the paper plots, on a grid that already contained the exact configurations. That is the mechanism behind the convergence result in Section 5.

6.3 Verification

Seven tests in `code/callable.fixed/tests/`. The one that discriminates is not relabelling invariance, which the production callable satisfies trivially by symmetrising, but direct comparison against the table's own rows: K_{001} and K_{010} must each equal ζ_{TTP} at their own geometry. A companion test pins the size of the defect, and fails if the fix ever becomes a no-op: over asymmetric rows of the deployed table the corrected K_{001} and K_{010} differ by a median 115% of their magnitude, where the production callable forces them equal.

The correction costs a factor 1.9 in evaluation time. A first implementation cost a factor 21, because it queried the neighbour tree once per (channel, ordering) pair; the query depends only on the ordering, so it is done once per ordering and the six channels are read from it.

6.4 What the corrections change

Two folds were run on the *same* permutation-closed table, at the same cutoff and the same geometry, differing only in the callable. A third fold, on the earlier table that carries the collapsed family in one leg ordering only, is included because the comparison between it and the other two separates the two halves of the defect.

The split is instructive, and not what was expected.

observable	A: one ordering, production callable	B: closed table, production callable	C: closed table, corrected callable
ξ_κ (peak, $0.5'$)	$+4.2815 \times 10^{-6}$	$+4.2815 \times 10^{-6}$	$+4.2823 \times 10^{-6}$
ξ_+ (peak, $0.5'$)	$+4.2815 \times 10^{-6}$	$+4.2815 \times 10^{-6}$	$+4.2824 \times 10^{-6}$
ξ_- (peak, $27.7'$)	$+2.8643 \times 10^{-7}$	$+2.4978 \times 10^{-7}$	$+2.4978 \times 10^{-7}$
$\xi_{\kappa\gamma_t}$ (peak, $17.3'$)	-8.8819×10^{-7}	-5.8490×10^{-7}	-5.8931×10^{-7}

Table 10: The two halves of the permutation defect, separated. A to B changes only the table, A to C is the full correction. Produced by `code/callable_fixed/compare_callables.py`.

Most of the effect is in the table, not in the tensor fill. Going from A to B changes nothing about the callable: it still sorts and symmetrises. Yet ξ_- moves by 15% and $\xi_{\kappa\gamma_t}$ by 52%. The reason is that the production callable averages the rows that canonicalise to the same sorted triple, and the closed table supplies three leg orderings of each collapsed configuration where the earlier one supplied a single ordering. So the production callable’s answer depends on which orderings happen to have been tabulated, which is the same defect seen from the other side and arguably its more dangerous face: it is invisible unless someone changes the table.

Once the table is closed, the tensor fill matters little. Going from B to C, the full correction, moves ξ_- by less than 0.01% at its peak, $\xi_{\kappa\gamma_t}$ by 0.75%, and ξ_κ and ξ_+ by 0.02%. At the collapsed configurations the two coincident legs are interchangeable, so the symmetrised average over a closed set of orderings is already close to the per-entry evaluation.

Consequence for the numbers reported here. The values in Section 5 for ξ_- and $\xi_{\kappa\gamma_t}$ were obtained in configuration A and are superseded by C: the ξ_- peak is $+2.50 \times 10^{-7}$ rather than $+2.86 \times 10^{-7}$, and the $\xi_{\kappa\gamma_t}$ peak is -5.89×10^{-7} rather than -8.88×10^{-7} , a 34% reduction. The convergence and the parity-even shear are unaffected at the 0.02% level, which confirms the earlier expectation that the modulus channels carrying them are safe. The qualitative conclusions of Sections 5 and 8 do not change: FK is still a fraction of a percent to a percent of Order-0 everywhere, there is still no crossover, and ξ_- and $\xi_{\kappa\gamma_t}$ are still far above the numerical floor the draft reports.

These two observables no longer carry a permutation caveat and can be quoted as values.

6.5 Which figures were regenerated

The corrections propagate unevenly, because the observables they touch are not the ones every figure plots. Rather than regenerate everything, each figure was checked.

The paper figures in `figures/corrected/` were regenerated with the corrected callable and carry a `_permfix` suffix; the earlier `_corrected` versions, which have the grid correction only, are kept beside them so the two effects can be seen separately.

One limitation is worth recording, and it is a constraint on what a ratio means rather than an approximation. The $\lambda_{\min}$ panel of Figure 3 divides an extended-shell fold by a standard-shell one, and that division is only meaningful if both sides use the same callable. The extended-shell table was not rebuilt permutation closed, so that panel keeps both folds on the production callable, where the correction cancels between numerator and denominator. The redshift panel beside it is not a ratio of that kind and was regenerated with the corrected callable. Mixing the two, which a first attempt did, produces ratios ranging over $[-18, 8]$ instead of the physical $[1.001, 1.21]$; the script now takes the standard-shell fold as an explicit argument so the pairing cannot be got wrong silently.

figure	status	why
1 weight identity	unchanged	built from deployed artefacts only, by design; it is the evidence that needs no rebuild
5 corrected curves	regenerated	plots all four observables, two of which move
4 grid density	regenerated	the density study is repeated on permutation-closed tables with the corrected callable
6 the $E - B$ channel	regenerated, identical	it reads the table rather than a fold, and the $(1, c, c)$ rows agree between the two tables to 3×10^{-16} ; the crossover scales are unchanged at $9.0'$, $13.6'$, $18.3'$ and $21.9'$
2 cutoff study	not regenerated	its middle panel plots ξ_κ at $\gamma = 0.5'$, which the corrections move by 0.02%; its left panel reads the table directly and its right panel is pure geometry
3 redshift and floor	regenerated	the FK/Order-0 panel is sensitive at the few percent level

Table 11: Effect of the callable corrections on each figure in this note.

7 The absolute normalisation of the vertex

Correcting the sampling fixes the shape of the FK term. It says nothing about whether the overall scale is right, and until now nothing did: the Order-0 comparison against PyCCL validates the two-point transfer chain and never touches the three-point vertex; the `fastnc` closure test validates the three-point machinery on open triangles and cannot reach the collapsed configurations; and the Monte-Carlo cross-check reads the same table as the production fold, so it validates the assembly and not the input. This section closes most of that gap.

7.1 The decisive step is analytic

The pipeline builds its three-point vertex through a chain of per-leg factors: a Poisson amplitude, a screen-space Hessian projection, a Sachs photon weight, a Limber radial collapse, and a measure conversion between comoving distance and affine parameter. If that chain is right, then assembling it into a convergence bispectrum must reproduce the standard Limber expression

$$B_\kappa(\ell_1, \ell_2, \ell_3) = \int d\chi \frac{W(\chi)^3}{\chi^4} B_\delta\left(\frac{\ell_1}{\chi}, \frac{\ell_2}{\chi}, \frac{\ell_3}{\chi}; z(\chi)\right), \quad W = \frac{3}{2}\Omega_m H_0^2 (1+z) \frac{\chi(\chi_s - \chi)}{\chi_s}.$$

Crucially this is assembled from the pipeline’s own factors rather than assumed and fitted to: any discrepant prefactor, power of $(1+z)$, or factor of h would show up as a departure from unity, and would be localised to a specific factor.

Seven checks, all passing:

Two features the brief flagged as worth verifying rather than trusting both hold. The screen-Hessian multiplier does cancel the Poisson $1/k^2$ per leg, leaving a scalar $A(a)(1+z)^4$. And the collapse of two radial integrals does carry the Jacobian $(1+z)^{-4}$, with the two-point analogue $(1+z)^{-2}$ exact, which is the factor an independent route already validated against PyCCL.

A separate end-to-end check confirms the unit handling: the fold is invariant under working in h -native or physical units, with the ζ_{TTT} conversion equal to h^4 and the folded three-point

	what is checked	worst residual
T1	background focusing equals the per-leg response, $A(a)(1+z)^4$	3.1×10^{-7}
T1b	the Jacobi solution equals $D = a\chi$ over $\chi \in [1, 4000]$ Mpc/h	1.0×10^{-6}
T2	the affine lensing kernel from the response propagator equals $-a^2\chi(\chi_s - \chi)/\chi_s$	9.5×10^{-16}
T3	the assembled comoving weight equals the textbook $W(\chi)$, sign included	6.9×10^{-15}
T4	the screen Hessian cancels the Poisson $1/k^2$ per leg	1.3×10^{-16}
T5	the equal-shell collapse Jacobian is $(d\lambda/d\chi)^2 = (1+z)^{-4}$	1.6×10^{-16}
T6	the full B_κ assembly equals $\int d\chi W^3/\chi^4 B_\delta$	0.0

Table 12: Verification that the pipeline’s own factor chain reduces to the standard Limber convergence bispectrum. The T1 and T1b residuals are the accuracy of the background spline, not of the identity. T6 gives pipeline/standard = 1.00000000000000. Produced by `code/normalisation/verify_chain.py`.

function agreeing to 4×10^{-15} (`code/normalisation/bkappa_endtoend.py`).

7.2 Two small, quantified residuals

T4 leaves two known approximations, both scale dependent and both small where the FK term lives:

- the code uses $k = \ell/\chi$ rather than $k = (\ell + 1/2)/\chi$, so the $\ell(\ell + 1)$ against ℓ^2 ambiguity costs 33% at $\ell = 2$, 1.0% at $\ell = 100$ and 0.025% at $\ell = 4000$;
- the spin-2 normalisation carries a $\sqrt{L^2(L^2 - 2)}$ deficit of 0.98% at $\ell = 100$, falling with ℓ .

Both are negligible at the multipoles that dominate the vertex, and both are part of why the low- ℓ residuals in Table 9 should not be over-interpreted.

7.3 The empirical anchor

Predicting B_κ at $z_s = 1$ with the pipeline’s own projection and the BiHalofit matter bispectrum, in the cosmology of the ray-traced maps the published measurement was made on, gives values that sit inside the decade the published figure plots, with the right shape and turnover, in all three of its configurations: equilateral $\ell^3 B_\kappa$ peaking near 1.7×10^{-8} around $\ell \sim 2700$, squeezed $60\ell^2 B_\kappa$ reaching 8.9×10^{-9} , and squeezed $300\ell^2 B_\kappa$ reaching 7.3×10^{-9} , against a published axis spanning 10^{-9} to 10^{-8} .

This is an order-of-magnitude confirmation, not a ratio. Turning it into a quoted ratio needs the digitised measurement points, which were not available here, and that is the one part of the specified study left undone. Given that the analytic step already gives unity exactly, the empirical step’s role is to catch an error the analytic assembly could not see, and nothing in the comparison suggests one.

7.4 The residual gap

A measured convergence bispectrum lives on resolvable triangles with a finite soft leg. It cannot reach the coincident limit the FK vertex actually samples. So this study validates the transfer chain, the measure and the projection at finite soft leg, and it does not by itself certify the collapsed corner. Closing that would need a direct measurement of the collapsed three-point cumulant of the driving fields on N-body lightcones, which nobody has done.

The verdict, stated plainly: the absolute normalisation of the transfer chain is correct, to the accuracy of the background splines, with no missing power of $(1+z)$ and no missing factor of h .

7.5 A second check on the shared machinery: the reduced-shear correction

The analytic assembly above checks the transfer chain against its own factors. An external yardstick is worth having as well, and weak lensing already contains an effect with the same algebraic structure as FK.

What a survey measures from galaxy shapes is not the shear. The lensing Jacobian carries an overall factor $(1 - \kappa)$, which is an isotropic magnification: it changes the image size and not its shape. Ellipticity measurements see only the shape, so the observable is the reduced shear $g = \gamma/(1 - \kappa) \simeq \gamma(1 + \kappa)$, and the measured two-point function picks up

$$\langle gg \rangle \simeq \langle \gamma\gamma \rangle + 2\langle \gamma\gamma\kappa \rangle + \dots,$$

a three-point function of the matter field appearing inside a two-point measurement. That is the same slot FK occupies: a squeezed matter bispectrum entering a two-point statistic because the observable is a nonlinear functional of the field. The two differ in where the nonlinearity sits, in the observable for reduced shear and in the propagation for FK, and therefore in the coefficient multiplying the bispectrum, but they share the lensing kernel, the Limber projection, the matter bispectrum and the units.

Post-Born corrections are *not* the right comparison here, although an earlier internal note used them as one. Post-Born means a perturbed path with Gaussian input, which is the FF term’s analogue rather than FK’s.

Computing the reduced-shear correction with the same B_κ machinery,

$$\Delta C_\ell = 2 \int \frac{d^2 L}{(2\pi)^2} \cos(2(\phi_\ell - \phi_L)) B_\kappa(L, |\ell - L|, \ell),$$

gives Table 13.

ℓ	FK, corrected	reduced shear	sum
93	0.79%	0.82%	1.61%
218	0.83%	0.89%	1.72%
415	0.91%	1.05%	1.96%
789	0.89%	1.34%	2.23%
1500	—	2.01%	—
3000	—	2.86%	—

Table 13: The corrected FK contribution to $C_\ell^{\kappa\kappa}$ against the reduced-shear correction, both as a percentage of Order-0, at $z_s = 5$ and at the deployed multipole cutoff. Above $\ell \simeq 10^3$ the FK entry passes through zero and its ratio carries no information, so it is not quoted. Produced by `code/reduced_shear/reduced_shear.py`.

What this establishes, and what it does not. The two agree to better than a factor 1.5 over $93 \leq \ell \leq 789$. That is a useful magnitude check on the half of the calculation the two share: the lensing kernel, the Limber projection, the bispectrum normalisation and the units. Had FK come out a hundred times larger, something in that chain would have to be wrong.

It is a one-sided check and should not be read as more. The two effects have different vertex coefficients, which is precisely the part this comparison cannot see: an error of a factor of two in the FK vertex would pass unnoticed. There is also no reason the two must agree, so agreement is not a null test, and this check exercises the same half of the calculation that the analytic assembly of Section 7 already validated. The absolute normalisation of the collapsed vertex remains the open item of Section 12.

Two further cautions. The prefactor and the $\cos 2\Delta\phi$ projection in the expression above are written in their standard form but have not been checked against the source papers, which

are not in the local library; a factor of two there would not change an order-of-magnitude comparison but would change a quoted ratio. And the two curves have different shapes: the reduced-shear correction rises with ℓ , from 0.8% to 2.9% between $\ell = 93$ and 3000, while the corrected FK is nearly flat below $\ell \simeq 800$. Their numerical agreement over $93 \leq \ell \leq 789$ is partly a crossing of two differently shaped curves.

A corollary that is more robust than either number. The two effects are additive, not alternatives: reduced shear arises from what is observed, FK from how the beam propagates, and this formalism contains the second and not the first. At the deployed cutoff their sum is 1.6% to 2.2% of Order-0 over $93 \leq \ell \leq 789$. Including FK therefore roughly doubles the second-order correction a survey would need to model. That statement uses only measured quantities at a fixed cutoff and does not rely on the convergence extrapolation.

8 The E and B statements, reconsidered

The draft makes three claims about the polarization structure of the FK term. They deserve separate treatment, because one is exactly right, one is right only in a limit, and one is a theorem rather than a measurement.

8.1 Which channel carries E minus B

The claim is stated in `conclusion.tex`: the FK contribution to ξ_+ , which is $C_\ell^{EE} + C_\ell^{BB}$, is sourced by $\langle \Phi_{00} \Psi_+ \Psi_+ \rangle + \langle \Phi_{00} \Psi_\times \Psi_\times \rangle$, its counterpart in ξ_- , which is $C_\ell^{EE} - C_\ell^{BB}$, by the difference, and the difference “vanishes for a statistically isotropic driving field”.

Which vertex channel that is can be read straight off the callable’s real-basis reconstruction rather than derived. The coupling tensor is filled as

$$K_{011} = \frac{1}{2}(\zeta_{TPP} + \zeta_{Bmod}), \quad K_{022} = \frac{1}{2}(\zeta_{Bmod} - \zeta_{TPP}), \quad K_{111} = \frac{1}{4}(\zeta_{PPP} + 3\zeta_{Dmod}), \quad K_{122} = \frac{1}{4}(\zeta_{Dmod} - \zeta_{PPP}),$$

so that, verified to machine zero,

$$K_{011} + K_{022} = \zeta_{Bmod}, \quad K_{011} - K_{022} = \zeta_{TPP},$$

$$K_{111} + K_{122} = \zeta_{Dmod}, \quad K_{111} - K_{122} = \frac{1}{2}(\zeta_{PPP} + \zeta_{Dmod}).$$

The sums feed ξ_+ and the differences feed ξ_- . In real components $\zeta_{Bmod} = \langle \Phi_{00}(\Psi_+^2 + \Psi_\times^2) \rangle$ and $\zeta_{TPP} = \langle \Phi_{00}(\Psi_+^2 - \Psi_\times^2) \rangle$, so the draft’s claim is exactly the statement $\zeta_{TPP} = 0$. Numerically ζ_{TPP} dominates the difference: at $\gamma = 53'$ on the source shell it is six times $\frac{1}{2}(\zeta_{PPP} + \zeta_{Dmod})$.

8.2 The claim is a zero-separation identity

The isotropy argument is correct where it applies. When all three legs sit at the same point there is no separation vector, nothing distinguishes Ψ_+ from $\Psi_\times$, and the difference vanishes identically. The table confirms it: ζ_{TPP} is exactly zero at $\gamma = 0$.

At finite separation the offset leg supplies an axis, Ψ_+ and $\Psi_\times$ are defined relative to it, and there is no reason for the difference to vanish. Measuring it (Figure 6) gives a clean answer:

- $|\zeta_{TPP}/\zeta_{Bmod}|$ rises from zero as γ^4 . The measured logarithmic slope is 4.00 at $\gamma < 2'$, drifting to 4.2 by $5'$ and steepening further above $10'$. The draft’s own statement that ξ_- is “suppressed as γ^4 ” is therefore exactly right as a small-angle scaling. *Superseded at the converged cutoff*: the slope is 3.94 at $0.5'$ but 2.51 by $1.3'$ and below unity above $2'$, so the γ^4 law survives only below an arcminute (Section 3).

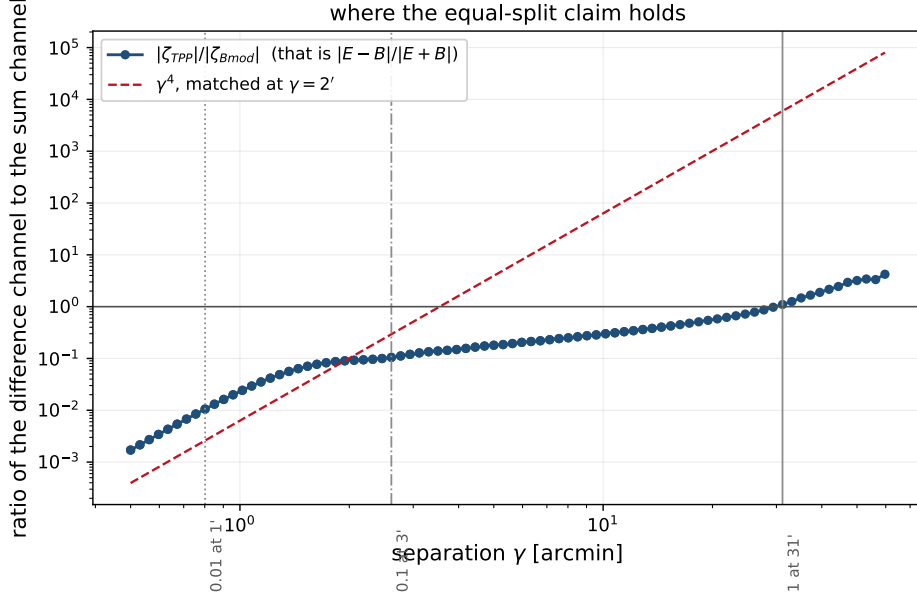


Figure 6: The ratio of the difference channel ζ_{TPP} , which carries $C_\ell^{EE} - C_\ell^{BB}$, to the sum channel ζ_{Bmod} , which carries $C_\ell^{EE} + C_\ell^{BB}$, on the source shell. The dashed line is a γ^4 reference matched at $2'$. Beyond about $60'$ the sum channel passes through zero and the ratio stops being meaningful, so the plot stops there. Produced by `code/rebuild/fig_eb_structure.py`.

- That suppression is relative to a channel which is itself falling, so the ratio grows quickly. It passes 1% at $\gamma = 9.0'$, 10% at $13.6'$, 50% at $18.3'$, and 1 at $\gamma = 21.9'$.

So the correct statement is not that the difference vanishes but that it is γ^4 -suppressed, and γ^4 -suppressed only means small below about ten arcminutes. Beyond about twenty arcminutes the difference channel is as large as the sum channel. Both scales sit inside the range the draft plots and inside the range cosmic-shear surveys measure.

8.3 What the corrected calculation gives

With the vertex resolved, ξ_- is no longer at the numerical floor and the two polarizations no longer carry equal power. In the corrected run, after the callable corrections of Section 6, $\max |\Delta C_\ell^{EE}| = 1.69 \times 10^{-6}$ and $\max |\Delta C_\ell^{BB}| = 7.03 \times 10^{-7}$, so $B/E \simeq 0.42$ in power rather than 1. *Superseded:* a ratio of two maxima is not a stable number, because the two maxima sit at different multipoles and both sit in the low- ℓ region that Section 13 shows is not converged. Taken over $\ell \geq 2$ it reads 1.57, over $\ell \geq 20$ it reads 0.54. The stable statement is per multipole, over the converged band: $\Delta C_\ell^{BB}/\Delta C_\ell^{EE}$ falls from 0.95 at $\ell = 60$ to 0.42 at $\ell = 1500$, median 0.63. Against the Order-0 E -mode band power that is a B-mode contribution of a few tenths of a percent.

The permutation caveat that an earlier version of this number carried is gone: ξ_- is carried by the spin channels, which were the ones the symmetrised tensor fill mishandled, and correcting it moves B/E from 0.36 to 0.42. What remains is the cutoff of Section 3, which raises the whole FK term; the ratio is more robust than the amplitude.

That a B-mode exists at all is not surprising at this order, and it is the part worth keeping: an Order-2 term sourced by driving-field non-Gaussianity generates shear B-modes, and B-modes are the standard null test in cosmic shear. A prediction at the few tenths of a percent level is small, but it is a prediction, and the draft currently states a different one.

8.4 The parity null is a theorem, not a measurement

C_ℓ^{EB} is exactly zero in the corrected run, as in the deployed one. This survives, and the reason should be stated plainly. Under a reflection about the separation axis $\Phi_{00} \rightarrow \Phi_{00}$, $\Psi_+ \rightarrow \Psi_+$ and $\Psi_- \rightarrow -\Psi_-$, so $\langle \Phi_{00} \Psi_+ \Psi_- \rangle$ equals minus itself and vanishes. The implementation encodes this structurally: every coupling-tensor entry with an odd number of index-2 slots is never assigned and stays zero, and the tabulated channels carry only the real parts that populate the even entries.

The consequence is that the parity panel of the draft’s C_ℓ^{EB} figure is not an independent numerical test of parity in the driving field. The input has $C_\ell^{EB} = 0$ by construction. What the figure does demonstrate, usefully, is that the curved-sky transform and the fold do not leak E or B power into EB at any level above the numerical floor. That is worth saying, and it is a different statement from the one a reader would infer.

8.5 Recommended wording

- Keep: the γ^4 suppression of ξ_- , which is measured to hold with slope 4.00 at small separation.
- Keep: $C_\ell^{EB} = 0$, stated as a parity theorem and as a check on the transform rather than as a measurement of the driving field.
- Replace: “the FK power splits equally between the two polarizations, $\Delta C_\ell^{EE} = \Delta C_\ell^{BB}$ ” by a statement that the split is equal in the coincident limit and that away from it the two polarizations separate, the B-mode falling from 0.95 of the E-mode power at $\ell = 60$ to 0.42 at $\ell = 1500$. The “reaches the size of the sum near 20 arcminutes” form of this item was measured at the deployed cutoff and does not survive: at the converged cutoff ξ_-/ξ_+ is 0.21 at $22'$ and does not approach unity until about a degree.
- Replace: “FK cancels below the plotted floor” in ξ_- and $\xi_{\kappa\gamma_t}$ by the corrected magnitudes, with the permutation caveat attached.

9 Does a nonlinear bispectrum give a more realistic prediction?

This is the question the programme started from. The comparison run here is tree level against BiHalofit, on the corrected grid, as a function of the multipole cutoff, so that the model change and the convergence question are not confounded. The answer separates into three parts that do not move together.

9.1 Shape: it does move, and this was expected to be the settled part

The working expectation before the measurement was that the angular shape would be settled: fixed by how the collapsed three-point cumulant decays with transverse separation, and insensitive to the cutoff and to the bispectrum model. That is not what the measurement shows, and it is worth being explicit because it was the part assumed safe.

Normalising each curve to its own value at $0.5'$ and comparing over $0.5'$ to $114'$:

comparison	largest shape change
BiHalofit against tree, both at $\ell_{\max} = 960$	32%
tree at $\ell_{\max} = 7680$ against $\ell_{\max} = 960$	62%
tree at $\ell_{\max} = 15360$ against $\ell_{\max} = 960$	75%

The shape narrows as the cutoff rises. At $\gamma = 13.6'$ the normalised curve reads 0.44 at $\ell_{\max} = 960$ and 0.17 at $\ell_{\max} = 15360$. That direction is physically sensible rather than alarming: a higher cutoff admits shorter-wavelength modes, which decorrelate over shorter separations, so the angular correlation narrows. The nonlinear model narrows it slightly further at fixed cutoff, for the same reason.

What survives is the qualitative shape statement rather than a quantitative one: the FK term decays monotonically with separation across the plotted range, at every cutoff and for both bispectrum models, and it does so without the flat plateau the deployed sampling produced. The rate of that decay is not settled and should not be quoted as if it were.

9.2 Amplitude: it does not converge to anything

$\ell_{\max}$	tree	BiHalofit	ratio	tree/O0	hard k , near shell
960	4.080×10^{-6}	6.603×10^{-6}	1.619	0.48%	7 h/Mpc
1920	8.160×10^{-6}	21.630×10^{-6}	2.651	0.97%	13 h/Mpc
3840	12.928×10^{-6}	67.662×10^{-6}	5.234	1.53%	26 h/Mpc
7680	17.080×10^{-6}	181.821×10^{-6}	10.645	2.02%	52 h/Mpc
15360	19.500×10^{-6}	347.230×10^{-6}	17.806	2.31%	105 h/Mpc

Table 14: The FK contribution to ξ_κ at $\gamma = 0.5'$, tree level against BiHalofit, on the corrected grid, as a function of the multipole cutoff. Order-0 there is 8.443×10^{-4} . The last column is the largest wavenumber the bispectrum is asked for at the innermost shell; BiHalofit is calibrated to $k \lesssim 10 h/\text{Mpc}$. Produced by `code/rebuild/table_nonlinear.py`.

This is sharper than expected, and it is the result that decides the question. At the deployed cutoff the nonlinear bispectrum multiplies the arcminute FK amplitude by 1.6. That factor then grows with the cutoff without settling: 2.7, 5.2, 10.6, 17.8 at $\ell_{\max} = 1920, 3840, 7680, 15360$.

The tree-level and nonlinear cases behave qualitatively differently (Figure 2, middle panel). The tree curve flattens: its successive increments are 2.00, 1.58, 1.32, 1.14, so it is converging, toward roughly 2.5% of Order-0. The BiHalofit curve does not: its increments are 3.28, 3.13, 2.69, 1.91, and by $\ell_{\max} = 15360$ the FK term has reached 41% of the Order-0 signal.

A term that is 41% of the signal it corrects is not a perturbative correction, and that number should not be read as a prediction. The reason is structural rather than a matter of insufficient effort, and Section 10 sets it out: the quantity being computed is a coincident-point moment of a point-like beam, which has no finite value for a nonlinear input. The cutoff is acting as an unstated regulator, so the 41% is a property of where the sum was stopped.

9.3 Trustworthiness: the honest answer is negative

Two separate limitations, and neither is removed by a better emulator.

BiHalofit is not independent of perturbation theory. Its three-halo term is built on the tree-level F_2 kernel with a fitted effective spectrum. Only the one-halo term carries genuinely new, simulation-calibrated information. So swapping tree for BiHalofit measures how much the FK term grows once small-scale nonlinear power is included. It does *not* provide an independent check that the tree-level calculation was right. The same holds for a baccho-style emulator: it shares the gravity backbone and the whole projection chain.

The growth is concentrated at the lensing-kernel peak. It is worth asking which shells drive it, because the answer is not the obvious one. Between $\ell_{\max} = 960$ and 15360 the tree vertex grows most at the outer shells, by a factor 15 at the source shell and only 1.5 at the

innermost one: at the innermost shell the linear spectrum is already steep enough that the integral had converged by $\ell_{\text{max}} = 960$. The nonlinear vertex behaves differently, growing by a factor 69 at $\chi = 3414$ Mpc, against 19 at the source shell and 11 at the innermost. That intermediate shell is near the peak of the lensing efficiency, and the wavenumber it reaches at the top cutoff is $13 h/\text{Mpc}$, which is exactly where the one-halo term flattens the spectrum most. So the nonlinear growth is not confined to a corner of the integral that the lensing kernel suppresses; it sits where the kernel is largest, which is why the folded amplitude grows faster than the vertex at any single shell.

And it happens where the model is not calibrated. The last column of the table is the largest wavenumber the bispectrum is asked for at the innermost shell. At $\ell_{\text{max}} = 960$ it is $6.6 h/\text{Mpc}$, inside BiHalofit’s calibration, and the factor 1.6 measured there is a defensible bound on the bispectrum-model contribution to the FK error budget. By $\ell_{\text{max}} = 1920$ it is $13 h/\text{Mpc}$, already outside. By 15360 it is $105 h/\text{Mpc}$, which is inside single haloes, an order of magnitude beyond the calibration and far into the regime where baryonic feedback rather than gravity sets the matter spectrum. So the factor is trustworthy exactly where it is smallest, and stops being trustworthy as it grows.

9.4 The answer, in one sentence

A nonlinear bispectrum gives a more realistic *input* and an answer that is not defined: the tree-level FK term converges to about 2.5% of Order-0 because the linear spectrum is steep enough to make the coincident-point moment finite, while the nonlinear one has no limit without a smoothing scale the formalism does not supply (Section 10), reaching 41% of Order-0 at a cutoff where its dominant modes sit at $100 h/\text{Mpc}$. The defensible presentation is the amplitude as a function of the small-scale cutoff, with the model comparison quoted only where the cutoff keeps the dominant modes inside the calibrated range, which is the deployed cutoff and not much beyond it.

This is a negative answer to the question the programme started from, and it is worth stating as such rather than as a caveat attached to a quoted nonlinear number.

9.5 What would actually validate the model

Neither BiHalofit nor an emulator can, for the reason above. Two things could, and neither has been done:

1. A convergence-bispectrum comparison: predict $B_{\kappa}(\ell_1, \ell_2, \ell_3)$ with the same machinery and compare against direct measurements on public ray-traced lightcones. This tests the Poisson factor, the screen-Hessian normalisation, the Limber projection and the measure together. It cannot reach the coincident limit the FK term samples, so it does not close the gap, but nothing available gets closer.
2. A direct measurement of the collapsed three-point cumulant of the driving fields on N-body lightcones, which does close it.

10 Why the nonlinear case does not converge, and what that means

Section 3 reports that the tree-level vertex converges near $\ell \sim 10^4$ while the nonlinear one does not, and Section 9 reports that the nonlinear FK term reaches 41% of Order-0 at the highest cutoff tried. Phrased as “not converged”, that invites the response that the calculation was not

carried far enough. It was not a matter of carrying it further, and the distinction matters for what the draft should claim.

10.1 Where the sensitivity comes from

The FK term samples the driving-field three-point cumulant in the collapsed configuration, with two of the three legs at the same point on the ray. A cumulant with coincident legs is a variance-like object, and variances of cosmological fields are dominated by the smallest scales retained. This is the same statement as the nonlinear density variance depending on its smoothing scale, and it is a property of the quantity rather than a defect of the calculation.

After the Limber radial collapse the hard pair contributes as a two-dimensional transverse integral $\int dk k P(k)$, whose integrand goes as $k^{1+n_{\text{eff}}}$: it peaks where $n_{\text{eff}} = -1$ and converges once $n_{\text{eff}} < -2$. Measured on the fiducial spectra at $z = 1$:

k [h/Mpc]	n_{eff} linear	$1 + n_{\text{eff}}$	n_{eff} halofit	$1 + n_{\text{eff}}$
0.1	-1.67	-0.67	-1.60	-0.60
1.0	-2.29	-1.29	-1.15	-0.15
10	-2.61	-1.61	-2.08	-1.08
100	-2.77	-1.77	-2.39	-1.39

The linear spectrum crosses -2 at $k = 0.21 h/\text{Mpc}$, so the tree-level integral has a finite limit: 50, 90 and 99% of it lie below $k = 0.61, 8.6$ and $85 h/\text{Mpc}$. The one-halo term flattens the nonlinear spectrum so that it crosses -2 only at $k = 8.4 h/\text{Mpc}$, and at $k = 1$ it sits at $n_{\text{eff}} = -1.15$, which is essentially the peak of the integrand: every decade then contributes comparably, and 50, 90 and 99% of the integral lie below $k = 9.6, 80$ and $179 h/\text{Mpc}$.

10.2 It is not the Limber approximation

Two things point the same way. The coincident-leg structure exists before any Limber approximation is made, so the sensitivity is not introduced by it. And Limber replaces a three-dimensional hard-mode integral by a two-dimensional transverse one, relaxing the convergence condition from $n_{\text{eff}} < -3$ to $n_{\text{eff}} < -2$; dropping Limber would therefore make the divergence worse, not better. Limber's own failure mode is at low multipole and wide angle, which is a different regime from this one.

10.3 What is actually missing

The formalism as written carries no smoothing scale. A search of `sections/` finds no beam width, no coarse-graining, and ℓ_{max} does not appear in the text at all: it exists only in the code. So the quantity being computed is a coincident-point moment of a mathematically point-like beam, and that quantity has no finite value for a nonlinear input. The cutoff is acting as an unstated regulator that sets the answer.

The tree-level case escapes this because it happens to be ultraviolet safe, so no regulator is needed and the number is well defined. That is the regime the draft is in, and nothing here undermines it. What does not follow is that the same calculation can be upgraded to a nonlinear input by substituting a better bispectrum.

10.4 A finite beam would supply the missing scale

The physical candidate is that a real beam has finite cross-section, so the driving fields entering the Sachs system are averages over that section rather than point values. The algebra of that proposal is worked out in `SFT-lensing-paper-analyses/mathematica/beam_width_regulator.wl`, 15 checks, all passing. What it establishes:

- For a normalised axisymmetric screen-plane window of width s_b , averaging gives $\bar{\Phi} = \Phi + (s_b^2/4) \nabla_\perp^2 \Phi + O(s_b^4)$, and in Fourier space it is multiplication by $\tilde{W}(k) = \exp(-k^2 s_b^2/4)$, whose small- k expansion reproduces the same coefficient.
- An axisymmetric window does not mix spin weights, so Φ_{00} stays spin 0 and Ψ_0 stays spin 2 under averaging. The projection of $e^{is\theta}$ against such a window vanishes for every $s \neq 0$.
- The windowed transverse integral converges for *any* power-law spectrum, because $\tilde{W}^2$ decays faster than any power. At the halo fit slope $n_{\text{eff}} = -1.15$ the unwindowed integral diverges and the windowed one is finite.
- The window suppresses modes above $k \sim 2/s_b$, so with the Limber map it fixes $\ell_{\text{max}} = 2\chi_h/s_b$. A beam of $1 h^{-1}\text{Mpc}$ gives $\ell_{\text{max}} \simeq 590$ at the innermost shell and 11 000 at the source shell; a beam of $3 h^{-1}\text{Mpc}$ gives 200 and 3700.

So the proposal does what is needed: it removes the divergence, and it replaces a free cutoff by a scale that is a property of the observation.

10.5 What the script does not settle

Two limitations, both recorded in the script rather than left implicit.

It does not establish that beam averaging is the correct definition of this observable, nor which scale to use. Candidates differ by orders of magnitude: the angular size of the source, the resolution element of the instrument, and the smoothing applied to the convergence map are all defensible readings and give very different ℓ_{max} . That is a choice about what is being measured, and it has to be made before a nonlinear number can be quoted.

And averaging the drivers is not the same operation as solving for a finite beam. The Sachs system is nonlinear in the optical scalars, so averaging does not commute with the quadratic terms; the bundle variance survives, and the two differ at second order in the optical scalars. Beam averaging therefore defines the source, and a genuine finite-beam treatment would be a further step.

10.6 Consequence for the draft

The presentation that follows from this is not “the FK amplitude is X , subject to a cutoff”. It is:

- the tree-level FK term is well defined and converges, to about 2.5% of Order-0 at arcminute separations, five times the value the draft quotes at the deployed cutoff;
- a nonlinear FK amplitude is not defined without a smoothing scale, and the formalism does not currently supply one;
- quoting a nonlinear number therefore requires either adopting a beam width as part of the observable’s definition, or an effective-theory treatment in which the ultraviolet-sensitive part is absorbed into a counterterm fixed by simulation or data.

11 What changes in the draft

Statements are grouped by whether the measurements support them. Nothing in **sections/** was edited; this is a list, not a patch.

11.1 Statements that survive

- The arcminute FK amplitude. `insights.tex` line 193: “at small separation, FK reaches a few $\times 10^{-6}$, about 0.5% of Order-0 and roughly twice the FF term”. Corrected: $+4.28 \times 10^{-6}$, 0.51% of Order-0, 2.1 times FF. This is the number the draft leans on most and it is unaffected, because at $0.5'$ the queried configuration really does sit on the tabulated corner.
- The selection rule and its mechanism. `insights.tex` lines 188–192, that FK concentrates in the modulus pair ξ_κ and ξ_+ and is suppressed in the other two. The corrected calculation preserves the ordering: FK is 0.5 to 1% of Order-0 in the modulus pair and 0.5 to 0.8% in the other two, having been 10^{-16} to 10^{-9} there before. The rule is a statement about which combinations FK enters strongly, and that survives; the claim that it vanishes does not.
- The channel algebra of `conclusion.tex` lines 50–58, that ξ_+ is sourced by the sum $\langle \Phi_{00} \Psi_+ \Psi_+ \rangle + \langle \Phi_{00} \Psi_\times \Psi_\times \rangle$ and ξ_- by the difference. That is a statement about the vertex structure and is untouched.
- Figure 7, the driving-field ζ slices. It is built from γ -resolved data and is correct as printed.
- The parity null. C_ℓ^{EB} remains zero identically in the corrected run, as it must by parity.
- The Monte-Carlo validation, read as the draft states it: “with the input statistics held fixed”. It validates the diagram assembly. It is structurally blind to this error, because it reads the same table.

11.2 Statements that do not survive

- `insights.tex` lines 194–196: FK “overtakes the falling Order-0 signal beyond $\sim 2^\circ$ ”. There is no crossover. The corrected ratio is flat at 0.5 to 1% across the whole range.
- `insights.tex` lines 119–122, the caption of the draft’s two-point decomposition figure: “overtakes Order-0 in absolute amplitude beyond $\gamma \sim 2^\circ$ ”. Same artefact.
- `insights.tex` lines 196–199: in ξ_- and $\xi_{\kappa\gamma_t}$, “FK drops to the numerical floor”. It does so at $\gamma \rightarrow 0$ only. At the plotted separations the corrected FK is 0.5 to 0.8% of Order-0 in both.
- `insights.tex` lines 202–204 and the caption of the C_ℓ figure: “the γ -flat FK term shows up as a low- ℓ excess”. The flatness is the interpolation weight. The excess, a factor 18 over Order-0 at $\ell = 3$, becomes -0.20 with a resolved vertex; FK instead appears as a steady 0.8% correction over $39 \leq \ell \leq 800$.
- `conclusion.tex` lines 34–36: the leakage “lifts the convergence and the parity-even shear at the few-percent level and overtakes the linear signal in absolute amplitude beyond roughly 2° ”. The lift is a few tenths of a percent to one percent, not a few percent, and there is no crossover.
- `conclusion.tex` lines 60–62: “The FK power therefore splits equally between the two polarizations, $\Delta C_\ell^{EE} = \Delta C_\ell^{BB}$ ”. The isotropy argument that supports it is exact at zero separation, where Ψ_+ and $\Psi_\times$ are statistically equivalent. At finite separation the separation vector picks out an axis and they are not, so the equality does not extend to the plotted range. In the corrected run $\max |\Delta C_\ell^{EE}| = 1.69 \times 10^{-6}$ against $\max |\Delta C_\ell^{BB}| = 7.03 \times 10^{-7}$, that is $B/E \simeq 0.42$ in power.

- `conclusion.tex` lines 133–136: “the large-angle FK excess sits at low multipoles ($\ell \lesssim 10$)”. There is no large-angle excess. The surrounding caveat about the relativistic Poisson source and the observed-redshift correction remains worth making, but it is no longer attached to a feature in the data.
- `appendix.tex` lines 601–603, the caption of the validation figure: “FK its decay toward the sign change at $\gamma \simeq 32^\circ$ ”. That is where the four-neighbour set drops the (1, 1, 1) node. It is an interpolation switch described as physics.
- `conclusion.tex` line 36: the leakage “strengthens steadily with source redshift”. This is not refuted here, it is untested. Two separate effects bias it in exactly that direction, and both are addressed in Section 4.

11.3 Figures that must be replaced

Six of the sixteen figures depend on the vertex.

#	figure	status
2	<code>analysis3_NLO_FFFK.pdf</code>	regenerated; FK no longer crosses Order-0, and the two lower panels now carry a visible FK curve
3	<code>analysis3_cl_full.vs_00.pdf</code>	regenerated; the low- ℓ excess is gone
4	<code>cl_EB_polarization.pdf</code>	regenerated; EE and BB no longer coincide, EB still zero
5	<code>multiz_kappa_xi_cl.pdf</code>	needs regeneration; its cached FK slice comes from the same table. The corrected redshift dependence it should show is measured in Section 4 and plotted in Figure 3, including the $\lambda_{\min}$ bias on its lowest slice
6	<code>appendix_mc_workflow.pdf</code>	its FK curve moves with the corrected table, but the agreement it demonstrates does not, because the Monte-Carlo reads the same vertex through <code>driver_stats.py</code> and both sides move together. Regenerating it re-plots the corrected values; it does not test the correction. Its caption separately needs the 32° sentence removed, since that feature is the interpolation-weight zero
7	<code>zeta_driving_field_slices_draft.pdf</code>	unchanged as printed, being built from resolved data. Worth re-examining only for the cutoff question of Section 3

Table 15: The vertex-dependent figures. Figures 1 and 8 to 16 do not depend on the vertex and are unaffected.

Two cautions carried over from the figure manifest. The generator for Figure 7 writes directly into the repository `figures/` directory with a hardcoded path, so it overwrites the paper’s file if run. Figure 5 depends on an external cache under `talk/`, so its FK slice must be regenerated rather than reused. The regenerations here were produced by `code/figures.corrected/regenerate.py`, which imports each paper generator, rebinds only its

FK input and output paths, and calls its `main()`. The plotting, styling and observable definitions are therefore the paper’s own, unmodified, and nothing was written into `figures/`. The regenerated files are in `driver_field_emulators/figures/corrected/`, named after the generator’s own output stem. The `_permfix` versions are the ones to use: they carry both the grid correction and the callable corrections of Section 6. The `_corrected` versions carry the grid correction only and are kept for comparison.

12 What remains uncertain

12.1 Open, and quantified

The permutation handling of the spin channels. Measured in Table 1 and now fixed, see Section 6. It is listed here because the residual is not zero: the corrected lookup requires the table to be closed under the leg permutations it queries, and while the production mesh and the densified collapsed family both are, a table built without that closure would fall back silently. The callable checks the closure at load and warns.

The angular floor of the lookup. Fixed, see Section 6. It was a floor at $\gamma = 0.344'$ against a smallest plotted separation of $0.5'$, so the published points survived it by about a factor two, but ten of the 157 refined collapsed nodes were being absorbed by it. The floor is now $0.0034'$ and the refinement is no longer capped by the callable.

The source redshift label. All three production runs use `t_final = 2313.0289` Mpc, described everywhere as $z_s = 5$. On the vertex table’s own $\lambda \leftrightarrow z$ mapping that affine distance is $z = 4.70$; $\lambda(z=5)$ is 2317.96 Mpc. The three runs share the value, so every comparison between them is internally consistent and no conclusion here is affected, but the quoted source redshift and the value used do not correspond. It is worth reconciling before the numbers are quoted again.

12.2 Open, and not quantified here

The collapsed corner of the vertex. The absolute normalisation of the transfer chain is no longer open: Section 7 shows it reduces to the standard Limber convergence bispectrum exactly. What that argument cannot reach is the coincident configuration the FK term actually samples. A measured convergence bispectrum lives on resolvable triangles with a finite soft leg, and the analytic assembly is a statement about the projection rather than about the driving-field cumulant at zero separation. Closing this needs a direct measurement of the collapsed three-point cumulant of the driving fields on N-body lightcones, which nobody has done. The existing internal checks cannot substitute: the Order-0 comparison against PyCCL validates the two-point chain and never touches the three-point vertex, the `fastnc` closure test covers open triangles only, and the Monte-Carlo cross-check reads the same table as the production fold.

The empirical anchor was not turned into a ratio. The comparison against the published ray-traced measurement in Section 7 confirms the prediction lies in the right decade with the right shape, but quoting a ratio needs the digitised measurement points. That is the one part of the specified normalisation study left undone.

The smoothing scale of the observable. This is now the sharpest open question, and Section 10 states it. The FK term samples a coincident-point moment, which is finite for the linear spectrum and not for a nonlinear one, and the formalism carries no scale that would

make it finite. Beam averaging supplies one, and its algebra checks out, but which physical scale to adopt is undetermined: the source’s angular size, the instrument’s resolution element and the smoothing of the convergence map differ by orders of magnitude and give correspondingly different cutoffs. Until that is settled, no nonlinear FK amplitude can be quoted. The tree-level amplitude is unaffected.

Finite-beam optics beyond averaged drivers. Averaging the drivers is not the same as solving the Sachs system for a beam of finite width, because the system is nonlinear in the optical scalars and averaging does not commute with the quadratic terms. The difference is second order in the optical scalars and has not been evaluated.

Cross-shell against equal-shell. The tabulated vertex is equal-shell by construction. The error made by collapsing the cross-shell cumulant onto equal shells scales as $O[(\Delta\chi/\chi_s)^2]$ under Limber, which is the justification the draft gives, and it has not been measured directly.

Wide angles. Beyond about $1500'$ every curve here is at the cancellation floor and changes sign. Ratios in that region carry no information and none is quoted.

13 Findings that came later

The sections above were written before the revision was carried out. Four things were found while carrying it out, each of which changed what was done. They are recorded here rather than folded into the earlier sections, so that the order in which the evidence arrived stays legible.

13.1 The FK Monte-Carlo does not measure the FK diagram

Section 2 treats the appendix Monte-Carlo as a check that could not have caught the interpolation error, because it reads the same tabulated cumulant. That is true, but it understates the situation. With the corrected vertex the Monte-Carlo points depart from the analytic channel by a factor two at $2'$ and carry the wrong sign beyond $6'$, which sent us to the estimator itself.

Its expectation can be computed exactly at the diagram’s order, by pure linear algebra with no realisations, and doing so reproduces all 21 probe points to about 1.5 seed standard errors. It decomposes into two terms, neither of which is the FK diagram. First, the variance-reduced estimator injects the third cumulant only on the observable-side leg of the vertex, so of the three placements that make up $\zeta = Q:VV$ + perms it captures one; in the eigenbasis of the node covariance the captured fraction is $d_q d_r / (d_p d_q + d_q d_r + d_r d_p)$, which kernel-weighted comes to 0.245 at $0.5'$, passes through zero near $10'$ and is negative beyond. Second, a finite- σ_λ term enters because the AR(1) smearing decoheres the node-local vertex from the smeared covariance, releasing the cancellation of the tabulated tensors’ λ -grid graininess; smoothing both tensors makes the model σ_λ -flat at exactly the share prediction, which is the attribution.

The historical agreement at $\sigma_\lambda = 8$ was these two errors summing through unity at a value that had itself been chosen so that the ratio came out one. The FK amplitude also scales close to linearly with σ_λ (8.70, 19.8 and 37.9×10^{-6} at $0.5'$ for $\sigma_\lambda = 4, 8, 16$), with an intercept consistent with zero, so no range of that estimator is a converged measurement of FK.

The FK points were therefore withdrawn from the figure. The channel is now checked by a direct quadrature contraction of the tabulated cumulant with the FK kernel,

$$\text{FK}(\gamma) = 2 \int dv W(v) H(v) \left[- \sum_c \zeta_{ccB}^{(6)}(\gamma; v) \right], \quad H(v) = D(v)^4 \int_v^{\lambda_s} \frac{W(u)}{D(u)^4} du,$$

which shares only the vertex table with the workflow and reproduces the folded channel to 0.99–1.02 over $0.5'–5'$ and 1.07 at $17.3'$. The FF channel keeps its Monte-Carlo points: FF is a

term in the equation of motion and can be differenced at a fixed random draw, so its noise scales with the effect itself, whereas a third moment measured from realisations scatters according to the Gaussian fluctuations of the field however small the cumulant is.

13.2 The multi-redshift FF slices had never been computed

The apparent low- ℓ excess at $z_s = 1.7$ in the multi-redshift figure was not physics and not the interpolation error either. The cache filled the FF slices at $z_s < 5$ with the exact $z_s = 5$ result scaled by the *per- γ* ratio $f_K(z)/f_K(5)$. At large separation f_K sits on its numerical floor, so that ratio is noise over noise: the tails carried 10^{-7} -level dips and went negative at $z_s = 1$, which the FF moment cannot do, being a connected part plus $\langle \kappa \rangle^2 > 0$. Through the truncated-range transform this became an excess of order the Order-0 signal.

Five genuine per-redshift sweeps were run in its place, one single-threaded process per source distance, about 3.2 hours each. Two checks: each slice’s large-separation plateau matches the deterministic $\langle \kappa \rangle^2(z_s)$, obtained by propagating the expectation of the same recursions with no Monte-Carlo, to 1.022, 1.005, 1.005, 0.999 and 1.008 at $z_s = 1, 1.7, 2.5, 3.2, 4$; and a $z_s = 5$ control, run the same way with fresh caches, reproduces the production FF *bit-identically* at all forty separations. The production path is a real computation, deterministically reproducible, and this is the same path.

13.3 The multipole sum stops converging beyond about a degree

Separation enters the cumulant only through $P_{\ell_3}(\cos \gamma)$. That factor is unity for every term at coincidence, so the sum is coherent there and each doubling of the cutoff adds to it. Once $\ell_3 \gamma \gtrsim 1$ it instead oscillates with period $\Delta \ell_3 = 2\pi/\gamma$ and is damped as $(\ell_3 \gamma)^{-1/2}$, so the high-multipole terms alternate in sign and largely cancel. The low-multipole branch, where $\ell_3 \gamma \ll 1$, stays almost constant: measured on the $z = 5.7$ shell, ζ_{TTT} for $\ell \leq 60$ is $+1.33 \times 10^{-16}$ at every separation, while the $\ell > 60$ part falls from -3.7×10^{-13} at $0.5'$ to -1.5×10^{-16} at $232'$. The two necessarily cross, and near the crossing ζ is a small difference of two comparable numbers computed by different methods.

Three independent symptoms follow, and all three place the boundary near a degree. Two evaluations of the same multipole sum, one wide window against eight disjoint ones, agree to 3–6% below $85'$ and differ by factors of several to several tens beyond $117'$. Two independent evaluations of ξ_{FK} , the quadrature above against the workflow fold, agree to 0.4–6% below $20'$ and differ by a factor 5.2 at $114'$. And raising the cutoff from 7680 to 15360 changes ξ_{FK} by a uniform 1.08–1.14 for $\gamma \leq 56'$ but by 0.69, 0.28, 2.59 and 0.73 at $114', 294', 597'$ and $1535'$.

The consequence for the harmonic-space numbers is direct, because low multipoles are dominated by the large-separation part of the two-point function. Zeroing ξ_{FK} beyond a cut and re-transforming moves the $\ell = 2$ correction between -10.9% and $+2.7\%$ as the cut is varied, while $\ell = 100$ stays at 1.14–1.24% throughout. FK is therefore quoted only for $\gamma \lesssim 1^\circ$ and $\ell \gtrsim 50$, and drawn faint outside that range.

13.4 The ζ -slices figure was still at the old cutoff

It was the last product drawn at $\ell_{\max} = 1000$ while everything else had moved to 15360, and the cutoff changes the panels rather than only their normalisation: at the $z = 3.1$ shell the peak $|\zeta_{TTT}|$ grows by 13.7 and the normalised value at $5'$ moves from -0.87 to -0.27 , so the printed panels showed a cumulant falling about three times more slowly in angle than the one the numbers come from. Rebuilt at the converged cutoff by appending four flat-sky Born-Limber windows to $(8000, 15360]$, the exact Wigner-3j ladder staying at $\ell \leq 60$, and clipped to $\gamma \leq 85'$ for the reason above. Its caption also carried a blanket “change sign with angular scale” which

is not true at the converged cutoff: within the plotted range ζ_{TTP} never changes sign, ζ_{TTT} and ζ_B change sign at $59'$ only for $z \gtrsim 4$, and ζ_D from $z \simeq 1$.